

Lesson 4: Understanding Internet Protocol

Multiple Choice

1. What is the default subnet mask for a Class B network?

- a) 255.0.0.0
- b) 255.255.0.0
- c) 255.255.255.0
- d) 255.255.255.255

Answer:

Difficulty: Easy

Section Reference: Categorizing IPv4 Addresses

Explanation: The IPv4 classification system is known as the *classful network architecture* and is broken down into five sections. Class B has a default subnet mask of 255.255.0.0.

2. You have a host with the address of 180.24.45.120. To what class would this be address be assigned?

- a) Class A
- b) Class B
- c) Class C
- d) Class D

Answer:

Difficulty: Medium

Section Reference: Categorizing IPv4 Addresses

Explanation: Class B begins with 128–191.

3. For IPv4, what class is used for multicasting?

- a) Class A
- b) Class B
- c) Class C
- d) Class D

Answer:

Difficulty: Medium

Section Reference: Categorizing IPv4 Addresses

Explanation: Class D is used for what is known as multicasting—transmitting data to multiple computers (or routers).

4. What is the maximum number of hosts you can use for each subnet that belongs to a class C address?

- a) 128
- b) 254
- c) 256
- d) 65,534

Answer:

Difficulty: Medium

Section Reference: Categorizing IPv4 Addresses

Explanation: A class C has 8 bits left for hosts. Therefore, 2^8 is equal to 256. But since you cannot use all 0s (network address) and all 1s (broadcast address), a class C will only support up to 254 hosts.

5. What type of address is 202.23.2.255?

- a) multicast address
- b) broadcast address
- c) unicast address
- d) anycast address

Answer:

Difficulty: Hard

Section Reference: Categorizing IPv4 Addresses

Explanation: The broadcast address is used to communicate with all hosts on the network. It is defined as all host bits are 0 (0000 0000).

6. What type of address is used to send a single set of packets to multiple hosts?

- a) multicast address
- b) broadcast address
- c) unicast address
- d) anycast address

Answer:

Difficulty: Medium

Section Reference: Categorizing IPv4 Addresses

Explanation: Multicasting addresses are used to transmit data to multiple computers (or routers).

7. Which of the following is not a private address?

- a) 10.4.24.24
- b) 172.16.36.45
- c) 192.168.34.3
- d) 200.23.34.120

Answer:

Difficulty: Medium

Section Reference: Categorizing IPv4 Addresses

Explanation: Private IP addresses are hidden from the Internet and any other networks. They are usually behind an IP proxy or firewall device. Several ranges of private IP addresses have been reserved by the IANA. Private address are 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.16.255.255 and 172.31.255.255 and 192.168.0.0 to 192.168.255.255.

8. What type of address is 169.254.32.23?

- a) APIPA
- b) multicast address
- c) anycast address
- d) broadcast address

Answer:

Difficulty: Easy

Section Reference: Categorizing IPv4 Addresses

Explanation: Microsoft developed one type of private range for use on small peer-to-peer Windows networks. This private range, called Automatic Private IP Addressing (APIPA), uses a single Class B network number: 169.254.0.0.

9. You have a computer that cannot connect to a server. When you look at the IP configuration, the host has an address of 169.32.54.2. What is the problem?

- a) The host cannot find a DHCP server.
- b) The host is set to multicast.
- c) The host is currently broadcasting.
- d) The host cannot find a domain controller.

Answer:

Difficulty: Easy

Section Reference:

Explanation: If a Windows client cannot get an IP address from a DHCP server and has not been configured statically, it auto-assigns an APIPA number on the network.

10. What defines the nearest router for a host?

- a) subnet mask
- b) default gateway
- c) DNS server
- d) WINS server

Answer:

Difficulty: Easy

Section Reference: Default Gateways and DNS Servers

Explanation: The default gateway is the first IP address of the device that a client computer will look for when attempting to gain access outside the local network. This device could be a router, server, or other similar device; it is the device that grants access to the Internet or other networks.

11. What server is used to translate host names to IP addresses?

- a) DNS
- b) WINS
- c) HOSTS
- d) DHCP

Answer:

Difficulty: Easy

Section Reference: Default Gateways and DNS Servers

Explanation: The DNS server address is the IP address of the device or server that resolves DNS addresses to IP addresses.

12. What is used to remap private addresses to public addresses and back?

- a) DNS
- b) DHCP
- c) WINS
- d) NAT

Answer:

Difficulty: Easy

Section Reference: Network Address Translation

Explanation: Network address translation (NAT) is the process of modifying an IP address while it is in transit across a router, computer, or similar device. This is usually so one larger address space (private) can be remapped to another address space, or perhaps remapped to a single public IP address.

13. What do you call a smaller network that is part of a larger network?

- a) host
- b) broadcast domain
- c) cluster
- d) subnet

Answer:

Difficulty: Easy

Section Reference: Subnetting

Explanation: Subnetting is the act of dividing a network into smaller logical subnetworks or subnets.

14. You have a subnet mask of 255.255.240.0. What would the CIDR notation be?

- a) /16
- b) /20
- c) /24
- d) /28

Answer:

Difficulty: Easy

Section Reference: Defining Classless Inter-Domain Routing (CIDR)

Explanation: One example of CIDR would be the IP network number 192.168.0.0/16. The /16 means that the subnet mask has 16 masked bits (or 1s), making 255.255.0.0. Usually, that would be a default Class B subnet mask, but because we are using it with what used to be a Class C network number, the whole address becomes classless.

16. How many bits is an IPv6 address?

- a) 32

- b) 48
- c) 64
- d) 128

Answer:

Difficulty: Easy

Section Reference: Understanding IPv6

Explanation: The top reason to use IPv6 is address space. IPv6 is a 128-bit system, whereas its still-dominant predecessor IPv4 is only a 32-bit system. While IPv4 can have approximately 4 billion IP addresses in the whole system, IPv6 can have 340 undecillion addresses.

17. What type of address is 2001:4212:0000:34DB:0000:0000:0000:4231?

- a) MAC address
- b) IPv4
- c) IPv6
- d) anycast address

Answer:

Difficulty: Easy

Section Reference: Understanding IPv6

Explanation: IPv6 addresses are also hexadecimal in format and divided into eight groups of four numbers each, with each group separated by a colon.

Fill in the Blank

18. A _____ address is assigned by a DHCP server.

Answer:

Difficulty: Easy

Section Reference: Categorizing IPv4 Addresses

Explanation: Most commonly, computers are set up to obtain an IP address (and other IP information) automatically. In this example of a dynamic IP address, the computer broadcasts out to the network in an attempt to find a DHCP server, whether it's a four-port SOHO router, DHCP server, or other appliance. The server then replies with the required information.

Short Answer

19. What is the loopback IP address?

Answer:

Difficulty: Easy

Section Reference: Categorizing IPv4 Addresses

Explanation: The range for Class A is 0–127. However, the 127 network number isn't used by hosts as a logical IP address. Instead, this network is used for loopback IP addresses, which allow for testing.

